

# Building a Simulated SOC: From Vulnerability Discovery to Intrusion Detection

(Video)

## 1. Executive Summary

This report documents detection of multiple vulnerabilities and simulated attacks against a deliberately vulnerable Windows VM. Using Kali Linux as the attacker and Ubuntu Snort IDS as the monitoring system.

Key findings include weak MySQL credentials, outdated Apache/PHP versions, and insecure SSL/TLS configurations. Snort successfully detected these attempts in real time, providing actionable alerts tied to known CVEs.

## 2. Objective

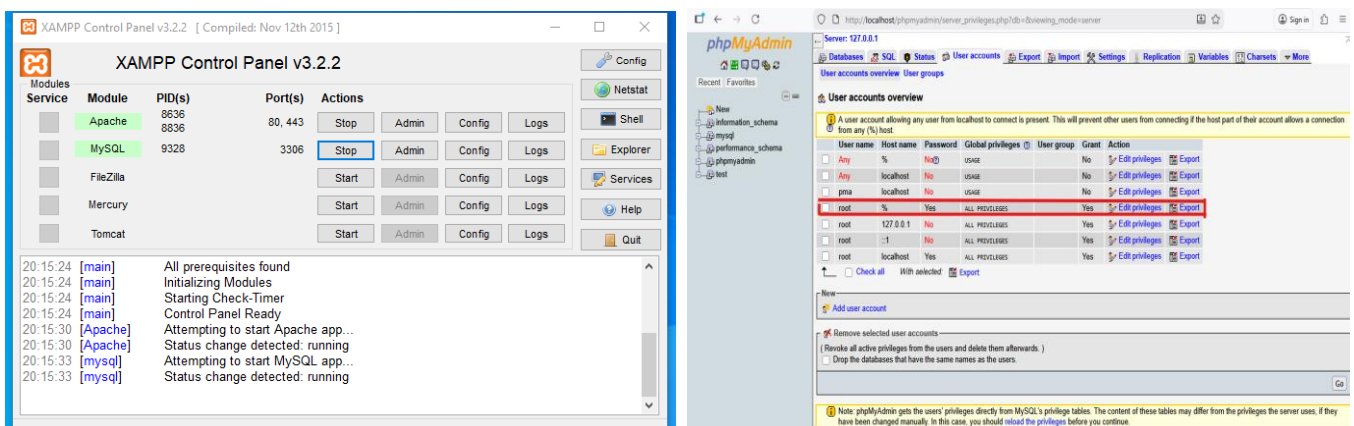
To build a simulated SOC environment that demonstrates vulnerability discovery, attack simulation, intrusion detection, and remediation using a three-VM setup:

- Windows VM (Victim) → vulnerable services (Apache, PHP, MySQL).
- Kali VM (Attacker) → Nessus, nmap, curl, mysql client, hydra.
- Ubuntu VM (SOC) → Snort IDS with custom rules.

## 3. Environment Setup

### Step A: Configure Windows VM (Victim)

- Installed XAMPP stack (Apache, PHP, MySQL/MariaDB).
- Set weak root password (root:1234).
- Left Apache and PHP at outdated versions:
  - ❖ Apache 2.4.29 (CVE-2017-15715).
  - ❖ PHP 7.2.0 (CVE-2019-11043, CVE-2020-7069).
- This created a deliberately vulnerable target for scanning and IDS testing.



## Step B: Kali VM (Attacker)

- Tools: Nessus Essentials, nmap, curl, mysql client, hydra used for scanning and simulating attacks.

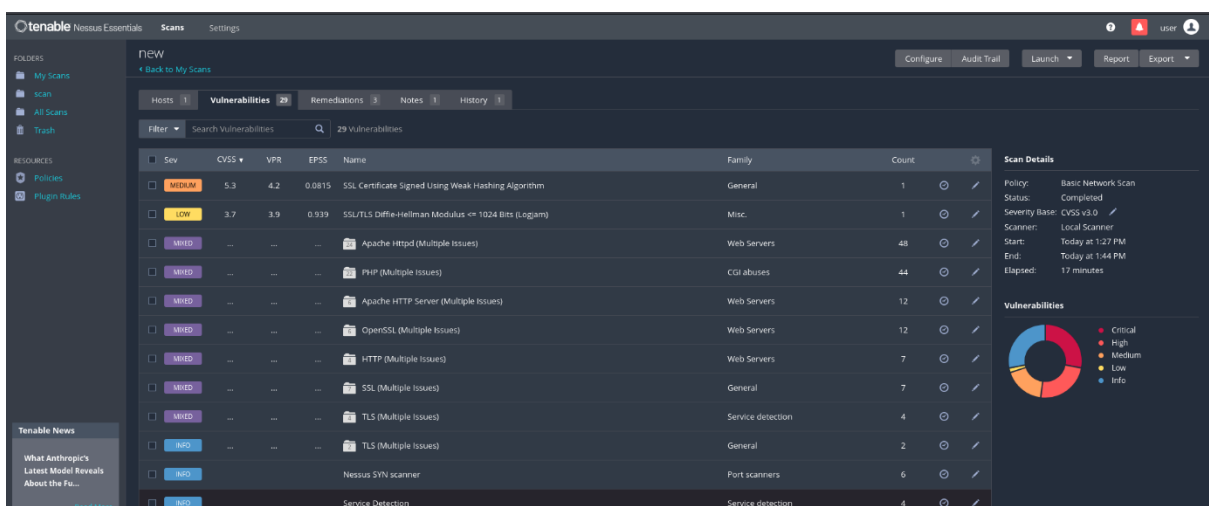
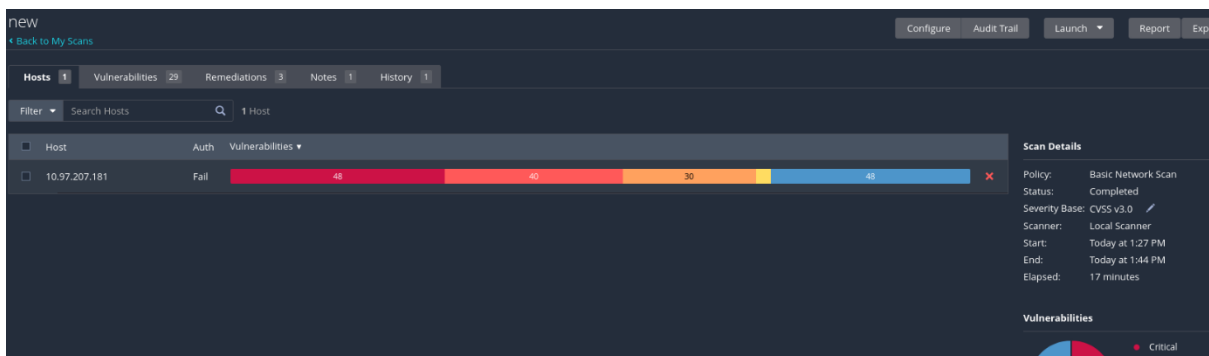
## Step C: Ubuntu VM (SOC Monitor)

- Installed Snort IDS.
- Custom rules added to `/etc/snort/rules/local.rules`.
- Alerts logged in `/var/log/snort/alert` and displayed live with `-A console`.

## 4. Workflow

### Step 1: Vulnerability Scanning (Kali)

- Ran Nessus against Windows VM.
- Detected multiple vulnerabilities:
  - ❖ Apache 2.4.29 (CVE-2017-15715).
  - ❖ PHP 7.2.0 (CVE-2019-11043, CVE-2020-7069).
  - ❖ MySQL weak authentication (CVE-2012-2122).
  - ❖ SSL/TLS weaknesses (OpenSSL CVEs).



scan / 10.97.207.181 / Apache Httpd (Multiple Issues)

Configure Audit Trail Launch Report

Vulnerabilities 29

Search Vulnerabilities 24 Vulnerabilities

| Sev      | CVSS | VPR | EPSS   | Name                                                      | Family      | Count |
|----------|------|-----|--------|-----------------------------------------------------------|-------------|-------|
| CRITICAL | 9.8  | 7.4 | 0.8709 | Apache 2.4.x >= 2.4.7 / < 2.4.52 Forward Proxy DoS / SSRF | Web Servers | 2     |
| CRITICAL | 9.8  | 7.4 | 0.8709 | Apache 2.4.x < 2.4.52 mod_lua Buffer Overflow             | Web Servers | 2     |
| CRITICAL | 9.8  | 7.4 | 0.4697 | Apache < 2.4.49 Multiple Vulnerabilities                  | Web Servers | 2     |
| CRITICAL | 9.8  | 6.7 | 0.9386 | Apache 2.4.x < 2.4.60 Multiple Vulnerabilities            | Web Servers | 2     |
| CRITICAL | 9.8  | 6.7 | 0.7582 | Apache 2.4.x < 2.4.46 Multiple Vulnerabilities            | Web Servers | 2     |
| CRITICAL | 9.8  | 6.7 | 0.6861 | Apache 2.4.x < 2.4.47 Multiple Vulnerabilities            | Web Servers | 2     |
| CRITICAL | 9.8  | 6.7 | 0.676  | Apache 2.4.x < 2.4.53 Multiple Vulnerabilities            | Web Servers | 2     |
| CRITICAL | 9.8  | 6.7 | 0.6704 | Apache 2.4.x < 2.4.56 Multiple Vulnerabilities            | Web Servers | 2     |

Scan Details

Policy: Basic Network Scan  
Status: Completed  
Severity Base: CVSS v3.0  
Scanner: Local Scanner  
Start: Today at 9:44 AM  
End: Today at 10:00 AM  
Elapsed: 16 minutes

Vulnerabilities

scan / 10.97.207.181 / PHP (Multiple Issues)

Configure Audit Trail Launch Report Export

Vulnerabilities 29

Search Vulnerabilities 22 Vulnerabilities

| Sev      | CVSS | VPR | EPSS   | Name                                                                               | Family     | Count |
|----------|------|-----|--------|------------------------------------------------------------------------------------|------------|-------|
| CRITICAL | 10.0 |     |        | PHP Unsupported Version Detection                                                  | CGI abuses | 2     |
| CRITICAL | 9.8  | 8.4 | 0.9411 | PHP < 7.1.33 / 7.2.x < 7.2.24 / 7.3.x < 7.3.11 Remote Code Execution Vulnerability | CGI abuses | 2     |
| CRITICAL | 9.8  | 6.7 | 0.8777 | PHP 7.2.x < 7.2.14 Multiple vulnerabilities                                        | CGI abuses | 2     |
| CRITICAL | 9.8  | 6.7 | 0.8307 | PHP 7.2.x < 7.2.3 Stack Buffer Overflow                                            | CGI abuses | 2     |
| CRITICAL | 9.8  | 6.7 | 0.5208 | PHP 7.2.x < 7.2.16 Multiple vulnerabilities                                        | CGI abuses | 2     |
| CRITICAL | 9.8  | 5.9 | 0.2029 | PHP 7.2.x < 7.2.8 Use After Free Arbitrary Code Execution in EXIF                  | CGI abuses | 2     |
| CRITICAL | 9.1  | 6.0 | 0.1005 | PHP 7.2.x < 7.2.19 Multiple Vulnerabilities                                        | CGI abuses | 2     |
| CRITICAL | 9.1  | 6.0 | 0.0447 | PHP 7.2.x < 7.2.17 Multiple vulnerabilities                                        | CGI abuses | 2     |

Scan Details

Policy: Basic Network Scan  
Status: Completed  
Severity Base: CVSS v3.0  
Scanner: Local Scanner  
Start: Today at 9:44 AM  
End: Today at 10:00 AM  
Elapsed: 16 minutes

Vulnerabilities

Tenable Nessus Essentials Scans Settings

Configure Audit Trail Launch Report Export

NEW

Back to My Scans

Hosts 1 Vulnerabilities 29 Remediations 3 Notes 1 History 1

Search Actions 3 Actions

| Action                                                                                                                               | Vulns | Hosts |
|--------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Apache 2.4.x < 2.4.64 Multiple Vulnerabilities: Upgrade to Apache version 2.4.64 or later.                                           | 83    | 1     |
| PHP 7.2 < 7.2.34 / 7.3.x < 7.3.23 / 7.4.x < 7.4.11 Multiple Vulnerabilities: Upgrade to PHP version 7.2.34, 7.3.23, 7.4.11 or later. | 42    | 1     |
| OpenSSL 1.1.0 < 1.1.0 Multiple Vulnerabilities: Upgrade to OpenSSL version 1.1.0 or later.                                           | 12    | 1     |

Scan Details

Policy: Basic Network Scan  
Status: Completed  
Severity Base: CVSS v3.0  
Scanner: Local Scanner  
Start: Today at 1:27 PM  
End: Today at 1:44 PM  
Elapsed: 17 minutes

## Step 2: Snort Rule Creation (Ubuntu)

- Wrote custom rules in *local.rules* based on Nessus findings.

```
GNU nano 7.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
#Apache
alert tcp any any -> 10.97.207.181 80 (msg:"Apache Httpd outdated version detected")

#PHP
alert tcp any any -> 10.97.207.181 80 (msg:"PHP 7.2.0 vulnerable version detected")

#MySQL Weak Credentials
alert tcp any any -> 10.97.207.181 3306 (msg:"MySQL weak credential attempt"; fast)

#SSL Traffic
alert tcp any any -> 10.97.207.181 443 (msg:"SSL traffic detected - check for weaknesses")

#OpenSSL Issues
alert tcp any any -> 10.97.207.181 443 (msg:"OpenSSL multiple vulnerabilities detected")
```

### Step 3: IDS Deployment

- Short running in IDS mode on Ubuntu.
- Monitoring traffic between Kali (attacker) and Windows (victim).
- Alerts configured to show in console and log files

```
user@Ubuntu:~/Desktop$ sudo snort -c /etc/snort/snort.conf -i enp0s3 -A console
[sudo] password for user:
Running in IDS mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830
2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777
7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300
8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002
55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 14
14 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:71
45 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 82
43 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444
41080 50002 55555 ]
```

### Step 4: Attack Simulation (Kali)

### 1) PHP Version Disclosure:

```
02/13-20:47:36.612184  ** [1:2152:1] WEB-PHP test.php access ** [Classification: access to a potentially vulnerable web application] [Priority: 2] {TCP} 10.97.207.190:37762 -> 10.97.207.181:80
```

```
02/13-20:47:36.612184  ** [1:100000121:1] COMMUNITY WEB-MISC Test Script Access ** [Classification: access to a potentially vulnerable web application] [Priority: 2] {TCP} 10.97.207.190:37762 -> 10.97.207.181:80
```

```
kali@kali:~/Desktop$ curl -I http://10.97.207.181/test.php
HTTP/1.1 404 Not Found
Date: Fri, 13 Feb 2026 15:17:37 GMT
Server: Apache/2.4.29 (win32) OpenSSL/1.1.0g PHP/7.2.0
Vary: accept-language,accept-charset
Accept-Ranges: bytes
Content-Type: text/html; charset=utf-8
Content-Language: en
```

## 2) SSL/TLS Traffic:

```
02/13-20:48:30.782222  [**] [1:200004:1] SSL traffic detected - check for weak c
erts [**] [Priority: 0] {TCP} 10.97.207.190:51982 -> 10.97.207.181:443
02/13-20:48:30.784469  [**] [1:200005:1] OpenSSL multiple vulnerabilities possib
le [**] [Priority: 0] {TCP} 10.97.207.190:51982 -> 10.97.207.181:443
02/13-20:48:30.784469  [**] [1:200004:1] SSL traffic detected - check for weak c
erts [**] [Priority: 0] {TCP} 10.97.207.190:51982 -> 10.97.207.181:443
```

```
02/13-20:50:03.350917 [**] [1:200006:1] Possible Nessus SYN scan [**] [Priority
: 0] {TCP} 10.97.207.190:37011 -> 10.97.207.181:1021
02/13-20:50:03.356298 [**] [1:200006:1] Possible Nessus SYN scan [**] [Priority
: 0] {TCP} 10.97.207.190:37011 -> 10.97.207.181:3689
02/13-20:50:03.358749 [**] [1:200006:1] Possible Nessus SYN scan [**] [Priority
: 0] {TCP} 10.97.207.190:37011 -> 10.97.207.181:5877
02/13-20:50:03.363591 [**] [1:200006:1] Possible Nessus SYN scan [**] [Priority
: 0] {TCP} 10.97.207.190:37011 -> 10.97.207.181:4900
02/13-20:50:03.363592 [**] [1:200006:1] Possible Nessus SYN scan [**] [Priority
: 0] {TCP} 10.97.207.190:37011 -> 10.97.207.181:37
02/13-20:50:03.363592 [**] [1:200006:1] Possible Nessus SYN scan [**] [Priority
: 0] {TCP} 10.97.207.190:37011 -> 10.97.207.181:6059
```

```
(kali@kali)~/Desktop$ nmap -ss 10.97.207.181
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-13 20:49 +0530
Nmap scan report for 10.97.207.181
Host is up (0.00085s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
3306/tcp  open  mysql
5357/tcp  open  wsdp1
MAC Address: 08:00:27:84:AE:54 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 5.59 seconds

(kali@kali)~/Desktop$
```

```
02/14:12:40:06.829548 [**] [1:2080083:1] MySQL weak credential attempt [**] [Priority: 0] [TCP] 10.97.207.190:56014 -> 10.97.207.181:3306
02/14:12:40:06.822455 [**] [1:2080083:1] MySQL weak credential attempt [**] [Priority: 0] [TCP] 10.97.207.190:56008 -> 10.97.207.181:3306
02/14:12:40:06.822459 [**] [1:2080083:1] MySQL weak credential attempt [**] [Priority: 0] [TCP] 10.97.207.190:56014 -> 10.97.207.181:3306
02/14:12:40:06.823193 [**] [1:2080083:1] MySQL weak credential attempt [**] [Priority: 0] [TCP] 10.97.207.190:56014 -> 10.97.207.181:3306
02/14:12:40:06.823196 [**] [1:2080083:1] MySQL weak credential attempt [**] [Priority: 0] [TCP] 10.97.207.190:56014 -> 10.97.207.181:3306
```

```
02/13-20:50:42.919836 [**] [1:20000:1] MySQL weak credential attempt [**] [Priority: 0] {TCP} 10.97.207.190:43848 -> 10.97.207.181:3306  
02/13-20:50:42.920808 [**] [1:20000:3:1] MySQL weak credential attempt [**] [Priority: 0] {TCP} 10.97.207.190:43848 -> 10.97.207.181:3306  
02/13-20:50:42.925012 [**] [1:20000:3:1] MySQL weak credential attempt [**] [Priority: 0] {TCP} 10.97.207.190:43848 -> 10.97.207.181:3306  
02/13-20:50:42.970595 [**] [1:20000:3:1] MySQL weak credential attempt [**] [Priority: 0] {TCP} 10.97.207.190:43848 -> 10.97.207.181:3306
```

```
(kali@kali)~/Desktop  
$ mysql --skip-ssl -h 10.97.207.181 -u root -p1234  
Welcome to the MariaDB monitor. Commands end with ; or \g.  
Your MariaDB connection id is 3  
Server version: 10.1.29-MariaDB mariadb.org binary distribution  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]>
```

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## 6) Remediation Plan

- ✚ Patch Management: Upgrade Apache, PHP, and MySQL to the latest stable versions to eliminate known CVEs.
- ✚ Access Control: Disable remote root login and enforce strong password policies across all services.
- ✚ Secure Communication: Configure MySQL to require SSL and harden SSL/TLS settings (restrict to TLS 1.2/1.3 only).
- ✚ Network Defense: Deploy firewall rules to block unauthorized scans and limit exposure of sensitive ports.
- ✚ Monitoring & Detection: Integrate Snort alerts into a SIEM platform for correlation, centralized visibility, and incident response.

## 7) References

1. Snort Documentation:  
<https://www.snort.org/documents>
2. Nessus Essentials User Guide:  
<https://docs.tenable.com/nessus>
3. Apache HTTP Server Project:  
<https://httpd.apache.org>
4. PHP Security Updates:  
<https://www.php.net/security>
5. MySQL Documentation:  
<https://dev.mysql.com/doc>
6. OpenSSL Project:  
<https://www.openssl.org/>

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